

Populating with random values

A DataFrame can be populated with random values by simply using the following code snippet:

```
df = DataFrame(rand(3,2))
print(df)
```

The output of the aforementioned code is:

```
3x2 DataFrame
| Row | x1      | x2      |
|     | Float64 | Float64 |
|-----|-----|
| 1   | 0.235304 | 0.664924 |
| 2   | 0.0916205 | 0.131976 |
| 3   | 0.136018 | 0.127914 |
```

Copying and comparison of DataFrames

A DataFrame object can be passed as an argument for building another DataFrame. Two DataFrames can be compared with `is_equal()`, as shown below:

```
df_one = DataFrame(a=1:5, b='p':'t')
df_two = DataFrame(df_one)
print(df_two)
isequal(df_one, df_two)
```

Here, `df_two` is built as a copy of the `df_one` DataFrame. Then, these two DataFrames are compared, and the output (true) is shown.

The output of this code is shown below:

```
5x2 DataFrame
| Row | a  | b  |
|     | Int64 | Char |
|-----|-----|
| 1   | 1  | 'p' |
| 2   | 2  | 'q' |
| 3   | 3  | 'r' |
| 4   | 4  | 's' |
| 5   | 5  | 't' |
```

```
true
```

3 rows × 8 columns

	variable	mean	min	median	max	nunique	nmissing	eltype
	Symbol	Union...	Any	Union...	Any	Union...	Union...	Type
1	one	1.5	1	1.5	2			Int64
2	two	5.0	5.0	5.0	5.0		1	Union{Missing, Float64}
3	three		x		y	2		String

Figure 2: `Describe()`: Summary statistics

10 rows × 8 columns

	x1	x2	x3	x4	x5	x6	x7	x8
	Int64	Int64	Int64	Int64	Int64	Int64	Int64	Int64
1	7	13	15	15	13	11	11	14
2	15	11	7	8	11	8	6	5
3	15	5	13	6	9	14	12	8
4	7	13	14	9	12	15	12	13
5	9	12	8	10	8	11	8	13
6	11	14	10	7	6	8	11	5
7	5	10	11	10	12	12	14	9
8	11	10	9	11	13	11	11	7
9	14	7	9	9	7	14	14	10
10	11	13	12	13	13	11	13	7

Figure 3: `First()` example

Getting summary statistics

Getting the summary statistics of the data populated in a DataFrame is done with the `describe()` function. The `describe()` function provides the following statistics:

- Mean
- Minimum
- Median
- Maximum
- Number of unique values
- Number of missing values
- Data type

For example:

```
df = DataFrame(one = [1, 2], two = [5.0, missing], three = ["x", "y"])
describe(df)
```

The output is shown in Figure 2.

In this example, `missing` represents the missing value. You can supply specific column names to restrict the columns for which the summary statistics would be built, as

shown below:

```
describe(df, cols=1:2)
```

Here, the summary statistics are built for the first two columns.

Selecting the first few rows

The first few rows of DataFrames can be selected using the `first()` function, as shown below:

```
df = DataFrame(rand(5:15, 5000, 8))
first(df, 10)
```

Here we are building a DataFrame with 5000 rows and eight columns. Each cell is filled with a random value between five and 15. Then, the first ten rows are selected with the `first(df,10)` function. The output is shown in Figure 3.

Similarly, the last few rows can be fetched with the `last()` function:

```
last(df, 10)
```

If the `first()` and `last()` are supplied without the second argument, then the first and last rows are displayed. Here, only one row will be displayed.

Fetching specific columns

DataFrames.jl provides so many different options to fetch specific columns. Covering all of them would be out of the scope of this article. You are encouraged to refer to the official documentation for the